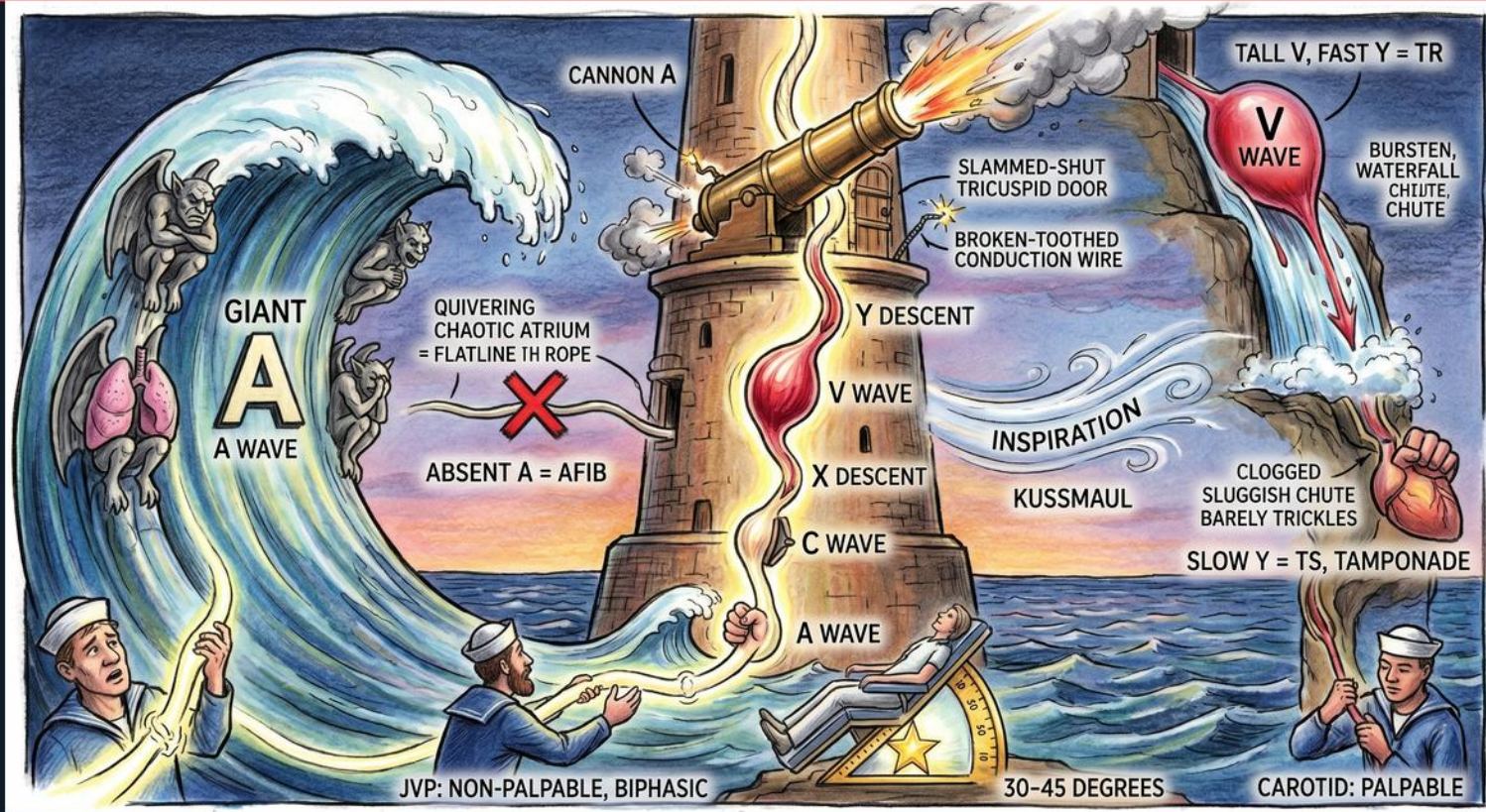
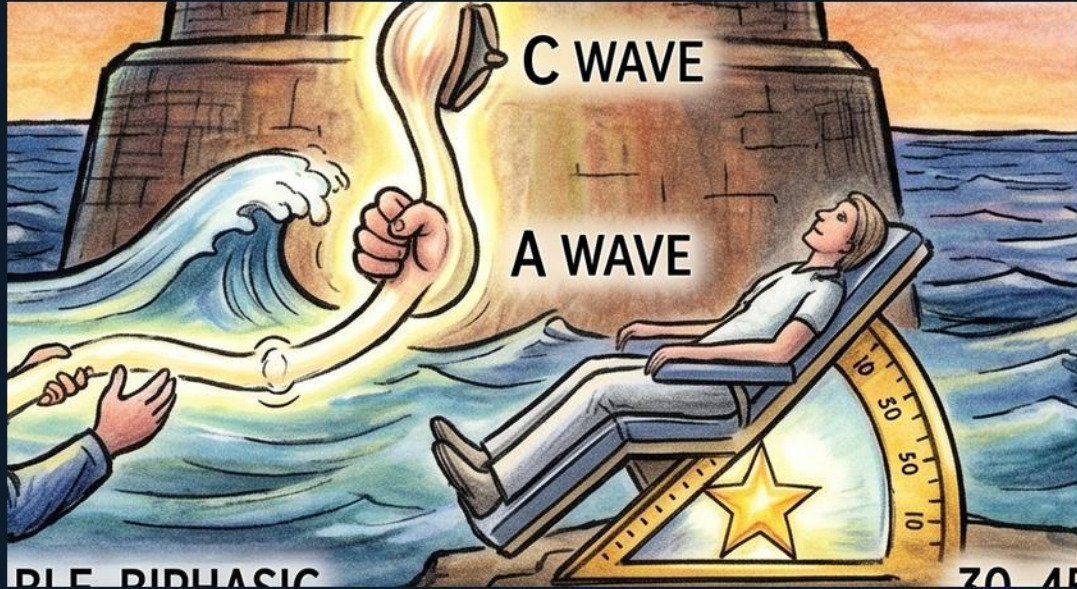


Cardiac Exam — JVP Waveforms & Signs





a wave

WHAT YOU SEE

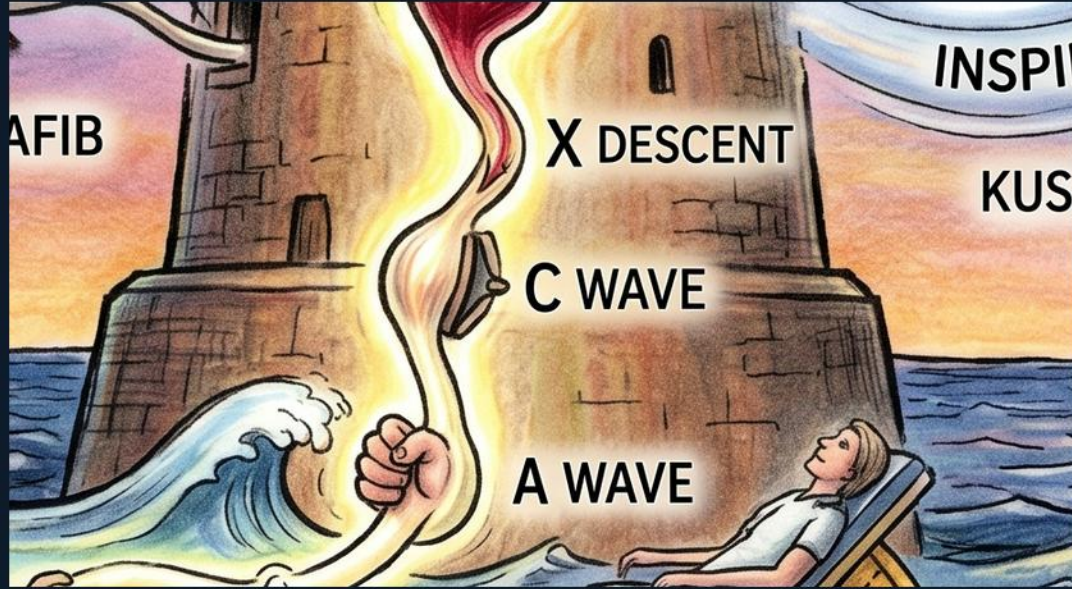
The lowest labeled bump on the central waveform column, tagged 'A WAVE'.

WHAT IT ENCODES

The a wave is the venous pulsation produced by RIGHT ATRIAL CONTRACTION in late diastole, just before S1. It corresponds to the P wave on ECG and is the first positive deflection of the JVP.

THE BOARD TRAP

Don't time it to systole — the a wave is PRESYSTOLIC (atrial kick), losing it points to loss of organized atrial activity.



c wave

WHAT YOU SEE

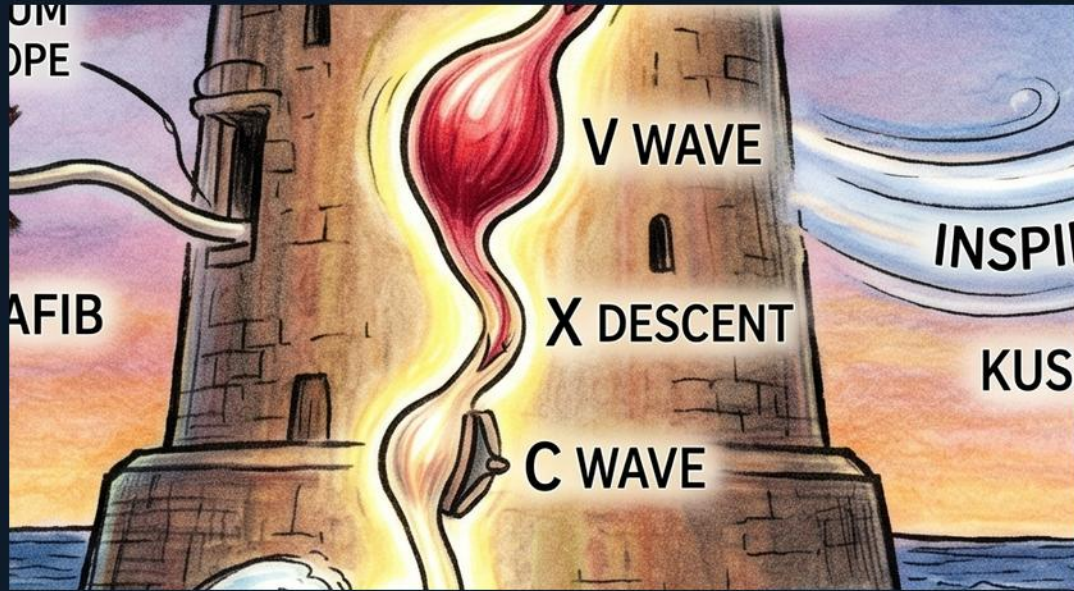
The bump labeled 'C WAVE' sitting just above the a wave on the column.

WHAT IT ENCODES

The c wave is the small bulging of the closed TRICUSPID VALVE into the right atrium at the very start of ventricular systole (isovolumetric contraction). It immediately follows S1.

THE BOARD TRAP

The c wave is a tricuspid bulge, not an atrial event — don't confuse it with the a wave's atrial contraction.



x descent

WHAT YOU SEE

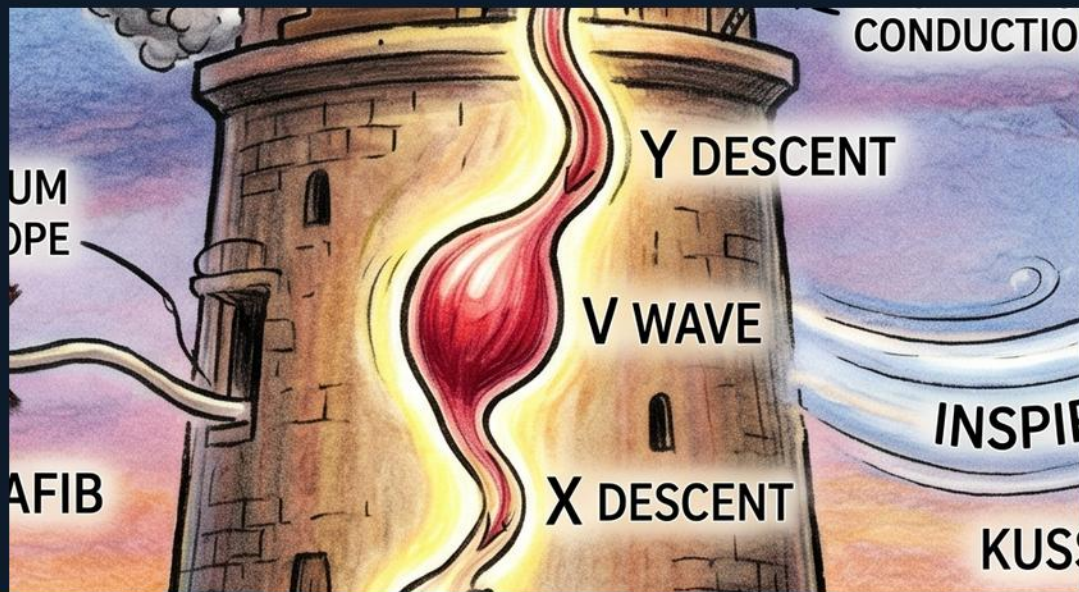
The downslope labeled 'X DESCENT' between the c wave and the v wave.

WHAT IT ENCODES

The x descent reflects ATRIAL RELAXATION plus downward pull of the closed tricuspid valve as the ventricle contracts. It is the dominant downward movement of normal JVP in systole.

THE BOARD TRAP

A blunted/absent x descent suggests tricuspid regurgitation — it gets replaced by the regurgitant systolic wave.



v wave

WHAT YOU SEE

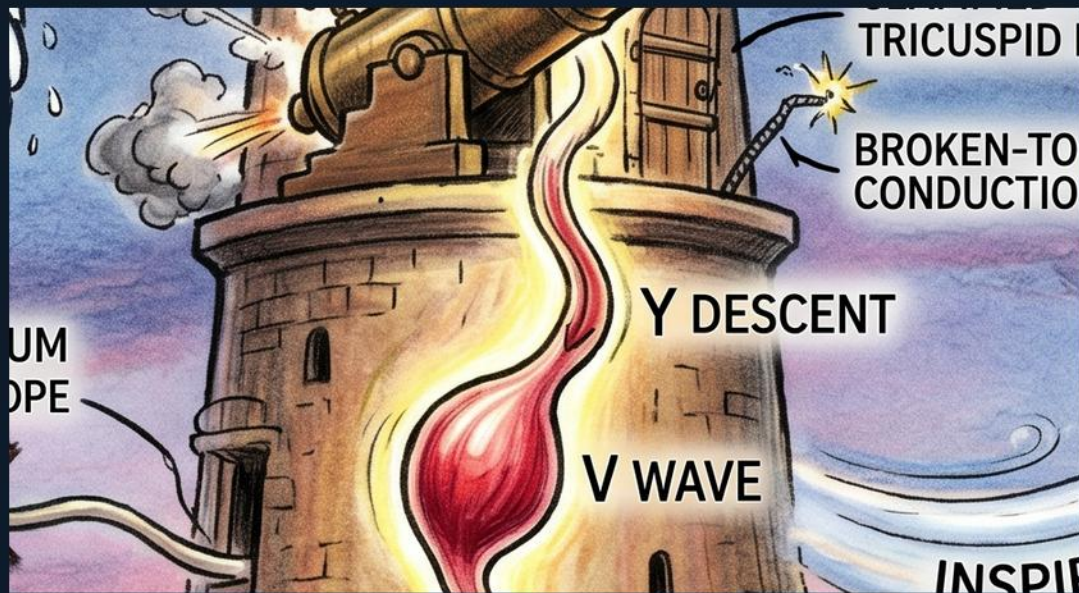
The bump labeled 'V WAVE' near the middle of the column.

WHAT IT ENCODES

The v wave is passive RIGHT ATRIAL FILLING against a CLOSED tricuspid valve during ventricular systole. It peaks at end-systole, just before the tricuspid opens.

THE BOARD TRAP

The v wave is atrial FILLING against a shut valve — not ventricular filling; a giant v wave means tricuspid regurgitation.



y descent

WHAT YOU SEE

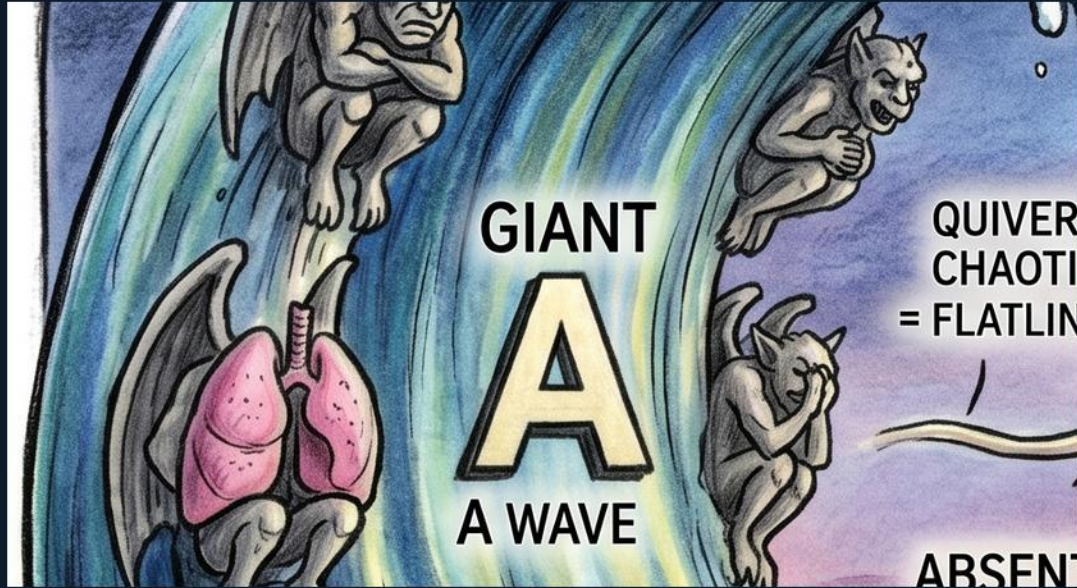
The downslope labeled 'Y DESCENT' high on the column, beneath the cannon.

WHAT IT ENCODES

The y descent is the fall in RA pressure as the TRICUSPID VALVE OPENS and the right atrium empties rapidly into the ventricle in early diastole.

THE BOARD TRAP

Slow/absent y descent = obstructed RA emptying (TS, tamponade, myxoma); a steep y descent = TR or constriction.



Giant a wave

WHAT YOU SEE

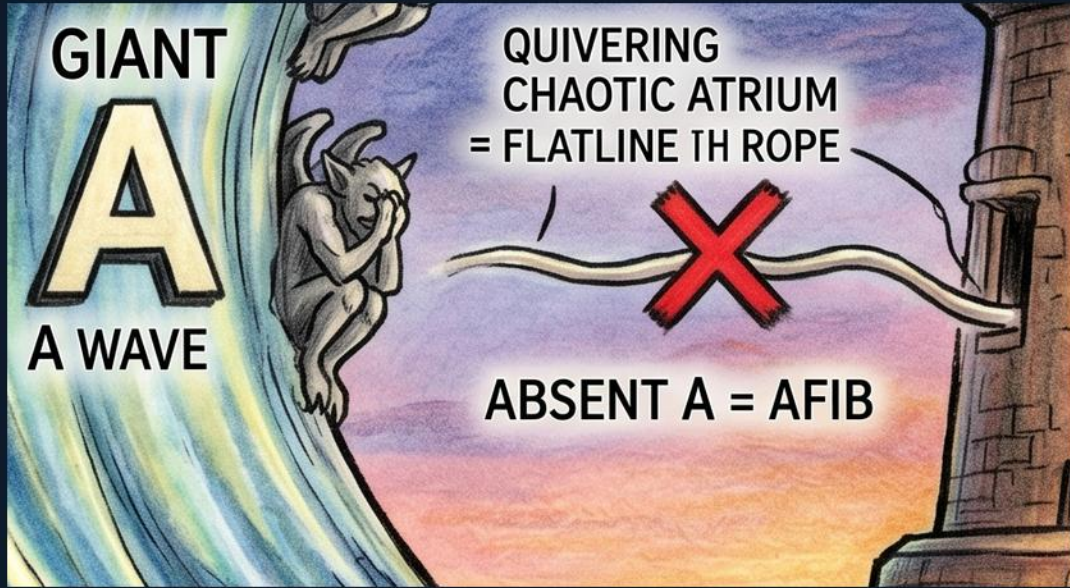
A huge towering 'A' wave on the left with gargoyles riding the crest and lungs/valve icons.

WHAT IT ENCODES

A GIANT a wave means the atrium contracts against high resistance to RA emptying: TRICUSPID STENOSIS, PULMONARY HYPERTENSION, or PULMONIC STENOSIS. The hypertrophied RA generates an exaggerated presystolic wave.

THE BOARD TRAP

Giant a is resistance to emptying (TS / pulm HTN / PS) — it still requires organized atrial contraction, so it cannot occur in atrial fibrillation.



Absent a wave = AFib

WHAT YOU SEE

A red 'X' over a quivering atrium with text 'ABSENT A = AFIB' and 'FLATLINE 1H ROPE'.

WHAT IT ENCODES

An ABSENT a wave indicates loss of organized atrial contraction — the hallmark is ATRIAL FIBRILLATION. With no coordinated atrial systole, the presystolic a wave disappears entirely.

THE BOARD TRAP

Absent a wave is pathologic (AFib), NOT a normal variant — recognizing its loss is the tested point.



Cannon a waves

WHAT YOU SEE

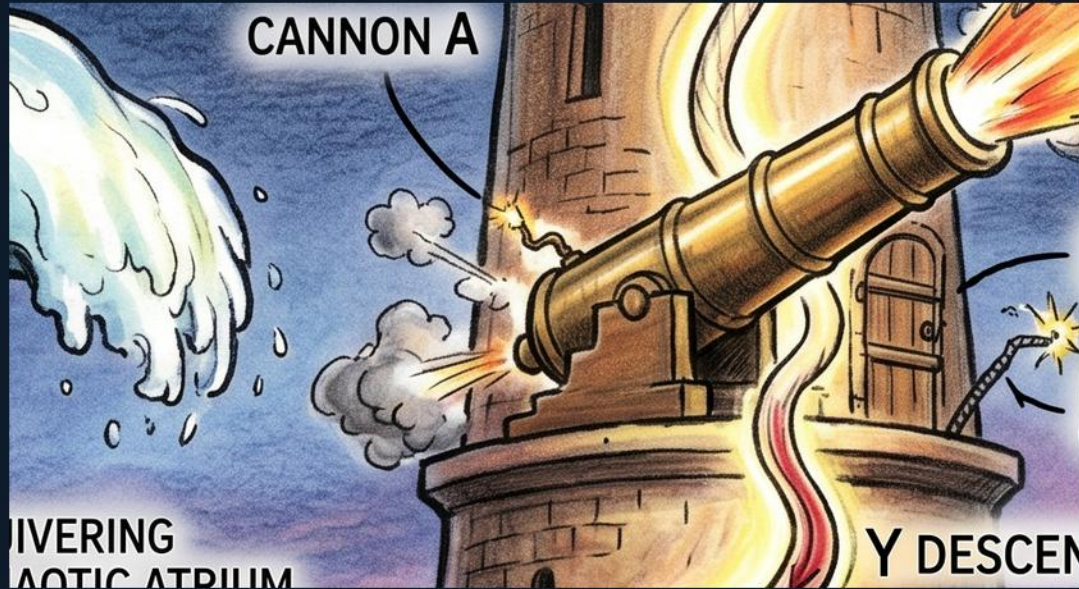
A firing cannon labeled 'CANNON A' on the lighthouse tower, blasting upward.

WHAT IT ENCODES

CANNON a waves occur in AV DISSOCIATION — the atrium contracts against a CLOSED tricuspid valve (ventricle already contracting). Causes: COMPLETE (third-degree) HEART BLOCK and VENTRICULAR TACHYCARDIA.

THE BOARD TRAP

Cannon a waves are AV dissociation, NOT a rhythm artifact — intermittent cannons are a bedside clue to complete heart block / VT.



Slammed-shut tricuspid

WHAT YOU SEE

Text 'SLAMMED-SHUT TRICUSPID DOOR' pointing at the cannon barrel.

WHAT IT ENCODES

The cannon fires because the atrium contracts when the tricuspid valve is already SHUT — atrial blood has nowhere to go but back up the jugular vein, producing the explosive wave.

THE BOARD TRAP

It is the closed valve at the wrong moment (dissociation), not a strong atrium per se, that makes the cannon.



Broken conduction wire

WHAT YOU SEE

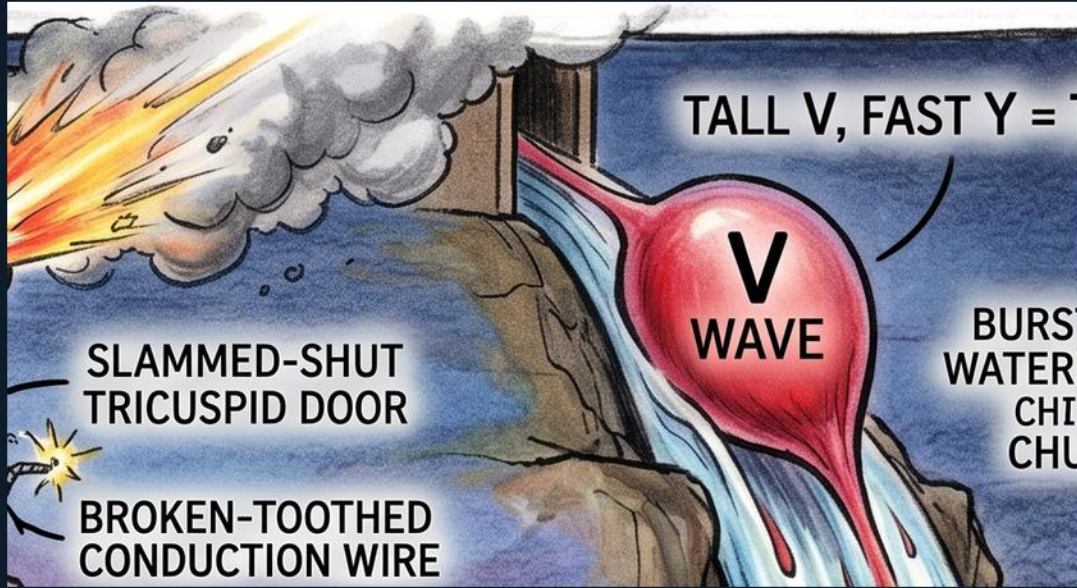
Text 'BROKEN-TOOTHED CONDUCTION WIRE' near the tower base.

WHAT IT ENCODES

The broken wire encodes AV CONDUCTION FAILURE — the electrical link between atria and ventricles is severed, so they beat independently (AV dissociation), which is the mechanism behind cannon a waves.

THE BOARD TRAP

AV dissociation is an electrical conduction problem; cannon a waves are its visible jugular consequence, not the cause.



Tall v + fast y = TR

WHAT YOU SEE

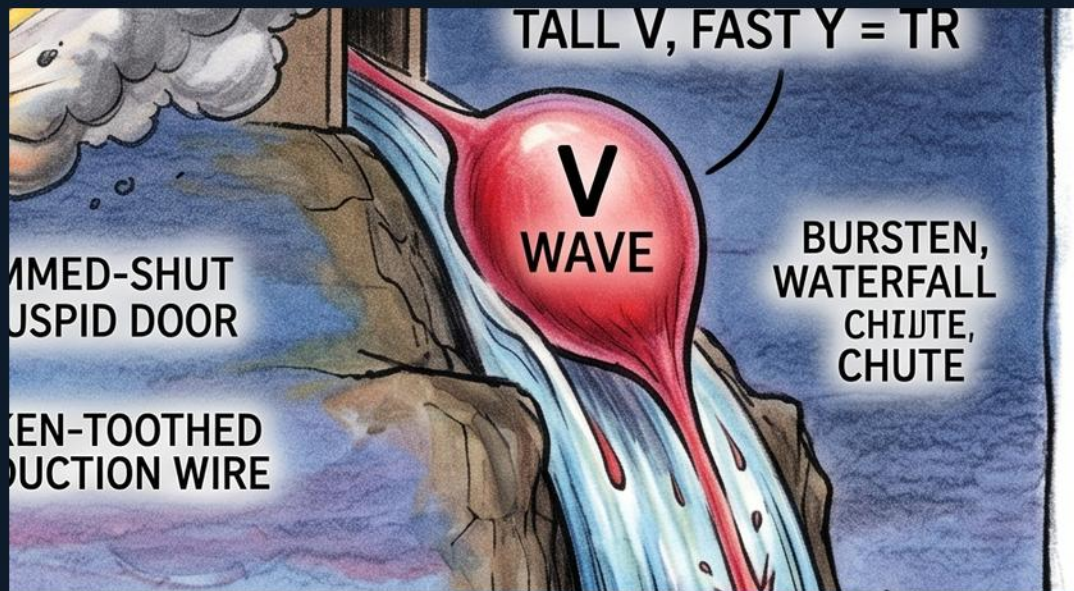
A red balloon-like 'V WAVE' top right labeled 'TALL V, FAST Y = TR'.

WHAT IT ENCODES

A TALL v wave with a RAPID y descent is the signature of TRICUSPID REGURGITATION — blood regurgitates into the RA during systole (giant v), then empties fast when the valve opens (steep y).

THE BOARD TRAP

This is TR, not TS — TR gives a TALL v with a STEEP/fast y, the opposite of TS's slow y.



Burst waterfall chute

WHAT YOU SEE

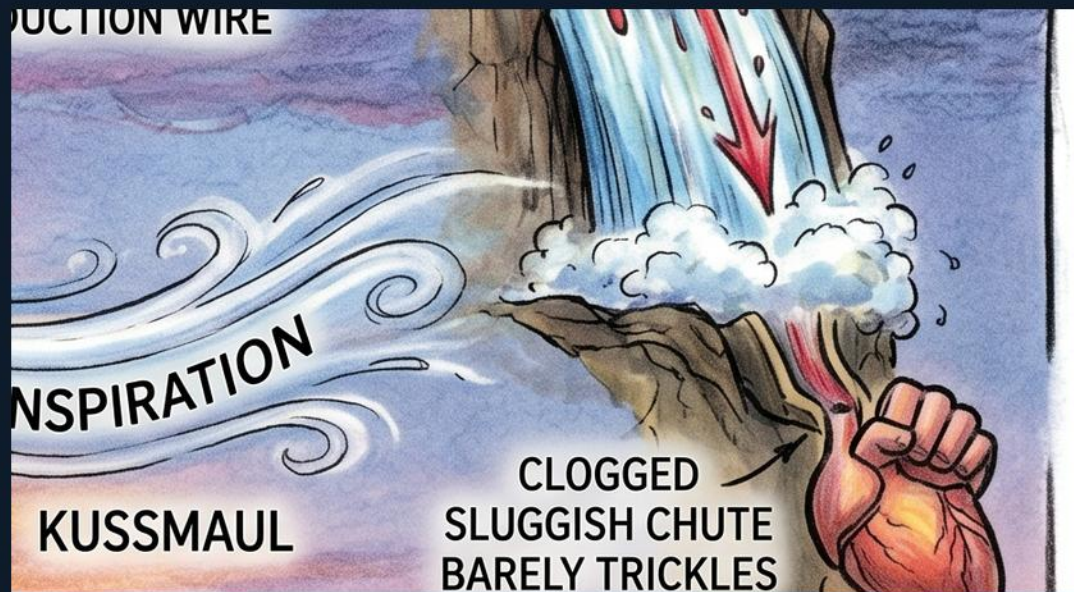
A bursting waterfall chute on the right labeled 'BURSTEN, WATERFALL CHUTE'.

WHAT IT ENCODES

The gushing waterfall depicts the rapid RA emptying of TR — the regurgitant volume drains quickly into the RV in early diastole, producing the brisk y descent.

THE BOARD TRAP

A fast, gushing y descent points to TR (or constriction), the opposite picture from the trickling slow-y of obstruction.



Slow y = TS / tamponade

WHAT YOU SEE

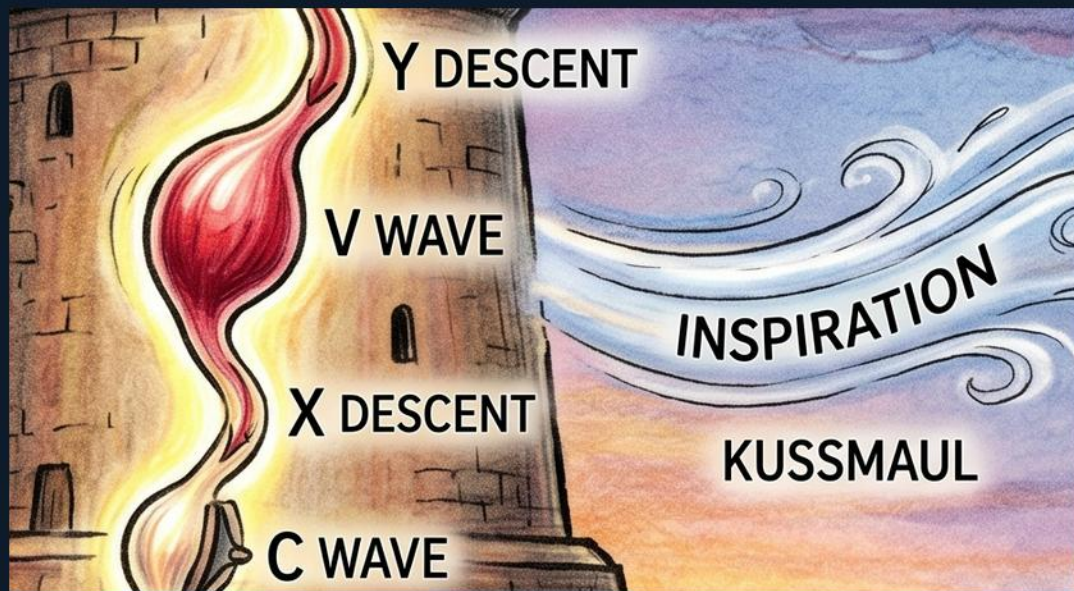
A clogged trickling chute labeled 'SLOW Y = TS, TAMPONADE' with 'BARELY TRICKLES'.

WHAT IT ENCODES

A SLOW or ABSENT y descent means impaired RA emptying: TRICUSPID STENOSIS, CARDIAC TAMPONADE, or RA MYXOMA. The RA cannot empty quickly, so the descent is sluggish.

THE BOARD TRAP

Slow y = obstruction to RA emptying (TS/tamponade/myxoma); don't confuse with the fast y of TR/constriction.



Kussmaul sign

WHAT YOU SEE

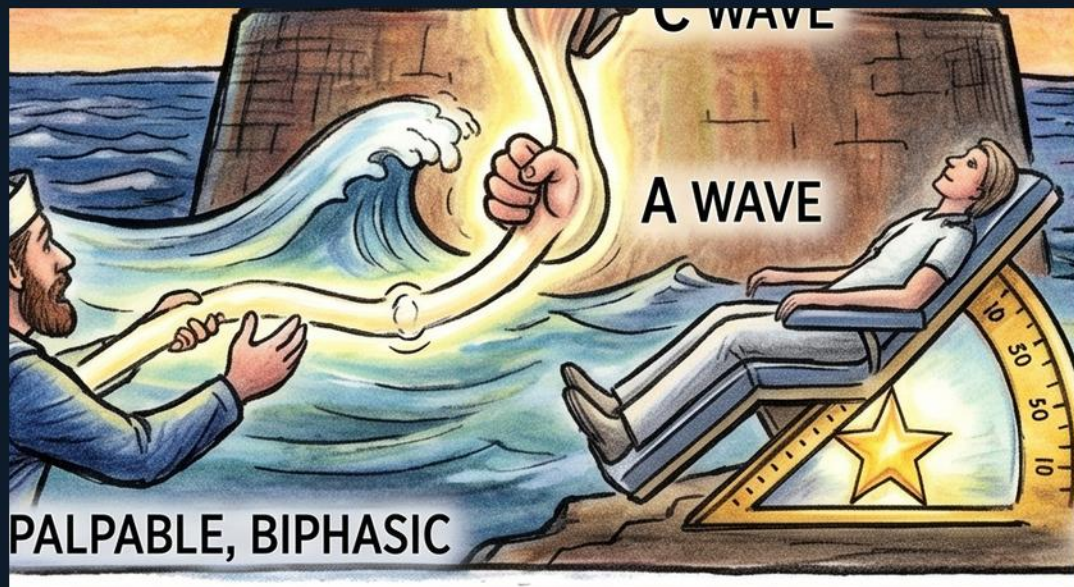
Wind labeled 'INSPIRATION' blowing over the word 'KUSSMAUL'.

WHAT IT ENCODES

Kussmaul sign is a PARADOXICAL RISE in JVP on INSPIRATION (normally JVP falls). It signals impaired RV filling: CONSTRICTIVE PERICARDITIS, RV INFARCTION, or RESTRICTIVE CARDIOMYOPATHY.

THE BOARD TRAP

Kussmaul is a paradoxical RISE on inspiration — normal JVP FALLS with inspiration; the rise is the abnormal, tested finding.



JVP vs carotid characteristics

WHAT YOU SEE

Sailors and footer text: 'JVP: NON-PALPABLE, BIPHASIC', '30–45 DEGREES', 'CAROTID: PALPABLE'.

WHAT IT ENCODES

JVP is NON-PALPABLE, BIPHASIC (two peaks per cycle), FALLS with inspiration and with sitting up, and is measured at 30–45°. The CAROTID is PALPABLE and monophasic — the key bedside distinction.

THE BOARD TRAP

Palpable = carotid, not JVP — a palpable, single-peak, non-respiratory pulsation is arterial, so don't read it as venous waveforms.

Summary — every symbol

1. a wave

The a wave is the venous pulsation produced by RIGHT ATRIAL CONTRACTION in late diastole, just before S1

3. x descent

The x descent reflects ATRIAL RELAXATION plus downward pull of the closed tricuspid valve as the ventricle contracts

5. y descent

The y descent is the fall in RA pressure as the TRICUSPID VALVE OPENS and the right atrium empties rapidly into the ventricle in early diastole.

7. Absent a wave = AFib

An ABSENT a wave indicates loss of organized atrial contraction — the hallmark is ATRIAL FIBRILLATION

9. Slammed-shut tricuspid

The cannon fires because the atrium contracts when the tricuspid valve is already SHUT — atrial blood has nowhere to go but back up the jugular vein, producing the explosive wave.

11. Tall v + fast y = TR

A TALL v wave with a RAPID y descent is the signature of TRICUSPID REGURGITATION — blood regurgitates into the RA during systole (giant v), then empties fast when the valve opens (steep y).

13. Slow y = TS / tamponade

A SLOW or ABSENT y descent means impaired RA emptying: TRICUSPID STENOSIS, CARDIAC TAMPONADE, or RA MYXOMA

15. JVP vs carotid characteristics

JVP is NON-PALPABLE, BIPHASIC (two peaks per cycle), FALLS with inspiration and with sitting up, and is measured at 30–45°

2. c wave

The c wave is the small bulging of the closed TRICUSPID VALVE into the right atrium at the very start of ventricular systole (isovolumetric contraction)

4. v wave

The v wave is passive RIGHT ATRIAL FILLING against a CLOSED tricuspid valve during ventricular systole

6. Giant a wave

A GIANT a wave means the atrium contracts against high resistance to RA emptying: TRICUSPID STENOSIS, PULMONARY HYPERTENSION, or PULMONIC STENOSIS

8. Cannon a waves

CANNON a waves occur in AV DISSOCIATION — the atrium contracts against a CLOSED tricuspid valve (ventricle already contracting)

10. Broken conduction wire

The broken wire encodes AV CONDUCTION FAILURE — the electrical link between atria and ventricles is severed, so they beat independently (AV dissociation), which is the mechanism behind cannon a waves.

12. Burst waterfall chute

The gushing waterfall depicts the rapid RA emptying of TR — the regurgitant volume drains quickly into the RV in early diastole, producing the brisk y descent.

14. Kussmaul sign

Kussmaul sign is a PARADOXICAL RISE in JVP on INSPIRATION (normally JVP falls)